Lecture 17

Taylor Expansion of Multivariable Functions

First Degree Taylor Polynomial: For a function of two variables $f(x,y)$ whose first partial derivatives exist at a point (a,b), the first degree Taylor polynomial of $f(x,y)$ near the point (a,b) is,

$$f(x,y) \approx L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

$L(x,y)$ is also called the linear approximation of $f(x,y)$ near the point (a,b).

Second Degree Taylor Polynomial: For a function of two variables $f(x,y)$ whose first and second partial derivatives exist at a point (a,b), the second degree Taylor polynomial of $f(x,y)$ near the point (a,b) is,

$$\begin{aligned} f(x,y) \approx Q(x,y) \quad &= f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b) \\ &+ \frac{f_{xx}(a,b)}{2!}(x-a)^2 + f_{xy}(a,b)(x-a)(y-b) + \frac{f_{yy}(a,b)}{2!}(y-b)^2 \end{aligned}$$

$Q(x,y)$ is also called the quadratic approximation of $f(x,y)$ near the point (a,b).

Example 1: Determine the first and second degree Taylor polynomial approximations for the function $f(x,y) = \sin 2x + \cos y$ near the point $(0,0)$.

Solution: Given, $f(x,y) = \sin 2x + \cos y$

and $(a,b) = (0,0)$

Here,

$f_x(x,y) = 2\cos 2x$

$f_y(x,y) = -\sin y$

$f_{xx}(x,y) = -4\sin 2x$

$f_{yy}(x,y) = -\cos y$

$f_{xy}(x,y) = 0$

So,

$f(0,0) = \sin 0 + \cos 0 = 0 + 1 = 1$

$f_x(0,0) = 2\cos 0 = 2 \cdot 1 = 2$

$f_y(0,0) = -\sin 0 = 0$

$f_{xx}(0,0) = -4\sin 0 = 0$

$f_{yy}(0,0) = -\cos 0 = -1$

$f_{xy}(0,0) = 0$

So, first degree Taylor's polynomial approximation,

$$L(x,y) = f(0,0) + f_x(0,0)(x-0) + f_y(0,0)(y-0)$$
$$= 1 + 2 \cdot x + 0 \cdot y$$
$$= 1 + 2x$$

Second degree Taylor's polynomial approximation,

$$\begin{aligned} Q(x,y) &= f(0,0) + f_x(0,0)(x-0) + f_y(0,0)(y-0) + \frac{f_{xx}(0,0)}{2!}(x-0)^2 + f_{xy}(0,0)(x-0)(y-0) \\ &+ \frac{f_{yy}(0,0)}{2!}(y-0)^2 \\ &= 1 + 2 \cdot x + 0 \cdot y + \frac{0}{2!}x^2 + 0 \cdot x \cdot y + \frac{-1}{2}y^2 \\ &= 1 + 2x - \frac{1}{2}y^2. \end{aligned}$$

Example 2: Determine the linear and quadratic Taylor expansion for $f(x,y) = x\sqrt{y}$ near (1,4). Also, find $f(1,2)$ using the expansion.

Solution: Given, $f(x,y) = x\sqrt{y}$

and $(a,b) = (1,4)$

Here,

$$f_x(x,y) = \sqrt{y}$$

$$f_y(x,y) = \frac{x}{2\sqrt{y}} = \frac{1}{2}xy^{-\frac{1}{2}}$$

$$f_{xx}(x,y) = 0$$

$$f_{yy}(x,y) = \frac{1}{4}xy^{-\frac{3}{2}} = -\frac{x}{4y^{3/2}}$$

$$f_{xy}(x,y) = \frac{1}{2\sqrt{y}}$$

So,

$$f(1,4) = 1 \cdot \sqrt{4} = 1 \cdot 2 = 2$$

$$f_x(1,4) = \sqrt{4} = 2$$

$$f_y(1,4) = \frac{1}{2\sqrt{4}} = \frac{1}{2 \cdot 2} = \frac{1}{4}$$

$f_{xx}(1,4) = 0$

$$f_{yy}(1,4) = -\frac{1}{4 \cdot 4^{\frac{3}{2}}} = -\frac{1}{4 \cdot 8} = -\frac{1}{32}$$

$$f_{xy}(1,4) = \frac{1}{2\sqrt{4}} = \frac{1}{2 \cdot 2} = \frac{1}{4}$$

Linear expansion,

$$\begin{aligned} L(x,y) &= f(1,4) + f_x(1,4)(x-1) + f_y(1,4)(y-4) \\ &= 2 + 2(x-1) + \frac{1}{4}(y-4) \\ &= 2 + 2x - 2 + \frac{y}{4} - 1 \\ &= -1 + 2x + \frac{y}{4}. \end{aligned}$$

Quadratic expansion,

$$\begin{aligned} Q(x,y) &= f(1,4) + f_x(1,4)(x-1) + f_y(1,4)(y-4) + \frac{f_{xx}(1,4)}{2!}(x-1)^2 \\ &\quad + f_{xy}(1,4)(x-1)(y-4) + \frac{f_{yy}(1,4)}{2!}(y-4)^2 \\ &= 2 + 2(x-1) + \frac{1}{4}(y-4) + \frac{0}{2!}(x-1)^2 + \frac{1}{4}(x-1)(y-4) + \frac{-\frac{1}{32}}{2!}(y-4)^2 \\ &= 2 + 2x - 2 + \frac{y}{4} - 1 + 0 + \frac{1}{4}(x-1)(y-4) - \frac{1}{64}(y-4)^2 \\ &= -1 + 2x + \frac{y}{4} + \frac{1}{4}(x-1)(y-4) - \frac{1}{64}(y-4)^2. \end{aligned}$$

Now, for the linear expansion,

$$\begin{aligned} f(1,2) &\approx L(1,2) \\ &= -1 + 2 \cdot 1 + \frac{2}{4} \\ &= -1 + 2 + \frac{1}{2} \\ &= \frac{3}{2}. \end{aligned}$$

For the quadratic expansion,

$$\begin{aligned} f(1,2) &\approx Q(1,2) = -1 + 2 \cdot 1 + \frac{2}{4} + \frac{1}{4}(1-1)(2-4) - \frac{1}{64}(2-4)^2 \\ &= -1 + 2 + \frac{1}{2} + \frac{1}{4} \cdot 0 \cdot (-2) - \frac{1}{64} \cdot 4 \\ &= \frac{11}{8}. \end{aligned}$$

Example 3: Determine the first and second degree polynomial approximations (Taylor Polynomials) for $f(x,y) = xe^y + 1$ near (1,0).

Solution: Given, $f(x,y) = xe^y + 1$ and

$$(a,b) = (1,0)$$

Here,

$f_x(x,y) = e^y$

$f_y(x,y) = xe^y$

$f_{xx}(x,y) = 0$

$f_{xy}(x,y) = e^y$

$f_{yy}(x,y) = xe^y$

Now,

$f(1,0) = 1 \cdot e^0 + 1 = 1 \cdot 1 + 1 = 2$

$f_x(1,0) = e^0 = 1$

$f_y(1,0) = 1 \cdot e^0 = 1 \cdot 1 = 1$

$f_{xx}(1,0) = 0$

$f_{xy}(1,0) = e^0 = 1$

$f_{yy}(1,0) = 1 \cdot e^0 = 1 \cdot 1 = 1$

So, first degree polynomial approximation,

$$\begin{aligned} L(x,y) &= f(1,0) + f_x(1,0)(x-1) + f_y(1,0)(y-0) \\ &= 2 + 1(x-1) + 1 \cdot y \\ &= 2 + x - 1 + y \\ &= x + y + 1 \end{aligned}$$

Second degree polynomial approximation,

$$\begin{aligned} Q(x,y) &= f(1,0) + f_x(1,0)(x-1) + f_y(1,0)(y-0) + \frac{f_{xx}(1,0)}{2!}(x-1)^2 \\ &\quad + f_{xy}(1,0)(x-1)(y-0) + \frac{f_{yy}(1,0)}{2!}(y-0)^2 \\ &= 2 + 1(x-1) + 1 \cdot y + \frac{0}{2!}(x-1)^2 + 1(x-1) \cdot y + \frac{1}{2!}y^2 \\ &= 2 + x - 1 + y + xy - y + \frac{1}{2}y^2 \\ &= x + xy + \frac{y^2}{2} + 1 \end{aligned}$$

1st degree = Linear = $L(x,y) = P_1(x,y) = T_1(x,y)$.

2nd degree = Quadratic = $Q(x,y) = P_2(x,y) = T_2(x,y)$.

H.W. Determine the linear and quadratic Taylor expansion for $f(x,y) = e^x \cos y$ at (0,0). Also find $f(1,2)$ using the expansions.

Answer: $L(x,y) = 1 + x,\ L(1,2) = 2.\ Q(x,y) = 1 + x + \frac{1}{2}(x^2 - y^2),\ \ Q(1,2) = \frac{1}{2}$.

H.W Determine the first and second degree Taylor's polynomial approximations for $f(x,y) = xe^y + 1$ in the powers of $(x-1)$ and $(y+1)$.

Answer: $L(x,y) = \left(\frac{1}{e} + 1\right) + \frac{1}{e}(x-1) + \frac{1}{e}(y+1)$.

$$Q(x,y) = \frac{1}{e}(x+y) + 1 + \frac{1}{e} + \frac{1}{e}(x-1)(y+1) + \frac{1}{2e}(y+1)^2.$$

H.W. Practice Sheet